Thanakrit Jomhong

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Education

Mahidol University Aug 2024 – Present
Bachelor of Engineering Program in Computer Engineering *GPA: 3.90*

Experience

Western Digital, Internship Jun 2026 – Aug 2026
Manufacturing Engineer (Machine Learning / Computer Vision), AVI Vision Enhancement

- Built a YOLOv8 instance segmentation pipeline for automated visual inspection across 6 defect classes, achieving 45.3% mask mAP@50 on 142 held-out test images and GPU inference at 7.7 ms/image.
- Constructed a labeled defect dataset by hand-annotating 1,500+ production images across 6 defect classes using open-source annotation tooling. Established labeling documentation standards to ensure annotation consistency and enable dataset expansion.
- Communicated project progress, experimental results, and technical challenges to the engineering mentor and manager through weekly reports and discussions.

Head of Sponsorship Coordination, ClickCamp16 Core Team May 2025 – Dec 2025
Mahidol Engineering, Computer Engineering Department

- Led sponsorship strategy and relationship management for a university computing camp; secured 130,000 THB in funding.
- Contacted 200+ companies and built structured outreach-tracking workflows to manage the full sponsorship pipeline.

Piscine, 42 Bangkok School May 2024 – Jun 2024
42 School, KMITL

- Completed the intensive one-month selection program, delivering peer-evaluated C projects under strict deadlines with no formal instruction.
- Core topics: C programming, UNIX and shell, Git and version control.

Selected Projects

Banana Ripeness Classification | ResNet34, PyTorch, scikit-learn

- Built a proprietary dataset from scratch by photographing bananas daily over 3+ months across their full ripeness lifecycle, capturing natural progression from fresh to rotten under real conditions.
- Iterated on class taxonomy through multiple relabeling experiments, testing different ripeness stage boundaries to maximize inter-class separability, improving F1-score to 87.5%.

HR Chatbot | Self-hosted n8n, Docker, Supabase(PostgreSQL), LLM, RAG

- Integrated the LINE Messaging API with an n8n POST webhook endpoint, containerizing the local environment with Docker Compose and configuring ngrok HTTPS tunneling for webhook testing.
- Built a RAG pipeline over HR policy documents with LLM-based intent extraction, handling multiple intent categories end-to-end.
- Developed PostgreSQL queries and JavaScript workflow logic for leave balance validation, request approval and cancellation, and Google Calendar integration, with interactive LINE Flex Messages.

EnergyPulse | Python, SQL, SQLite, dbt, Apache Airflow, BigQuery

- Developed a data ingestion prototype processing 100,000 synthetic energy records in 0.194 seconds in a local in-memory SQLite benchmark.
- Implemented primary-key-based idempotent replay, reprocessing 10,000 existing records with zero additional rows in the prototype experiment.
- Evaluated SQL query optimization through indexed versus non-indexed execution, measuring a 99.9% reduction in median query latency for a selective analytical query.

Skills

Languages: Python, C, C++, Java, SQL, Rust, HTML / CSS
ML / CV: PyTorch, Ultralytics YOLO, OpenCV, scikit-learn, Pandas, NumPy
Data Engineering: ETL & ELT, Data Mining, Data Pipelines, Data Quality
Systems & Tools: Docker, Git/GitHub, FastAPI, Linux, Supabase, Vercel, n8n
English: TOEIC 865 / 990 (CEFR B2)

Awards

2nd Place, Practice AI, Super AI Engineer SS5 (AI Innovator Track, Western Region) Summer 2025